ENGR 213: Applied Ordinary Differential Equations

Lecture 12: Nonhomogeneous Linear Equations & The Method of Undetermined Coefficients

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Curriculum Reference: Advanced Engineering Mathematics (D.G. Zill, 7th Edition, Section 3.4)
Lecture Grounding: Dr. A. Haghighat M., Lecture 12 (October 23, 2026)

1. Executive Conceptual Intuition & Physical Analogy

> [!IMPORTANT] **Core Concept Overview**: In Lecture 11, the right-hand side of our differential equation was 0 (a system left alone to decay or oscillate based solely on its internal parameters). In Lecture 12, an external driving force $g(x)$ acts on the system:
>
> $$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = g(x)$$
>
> By the **Superposition Principle**, the total response of the system is the sum of two independent components:
>
> $$y(x) = y_c(x) + y_p(x)$$
>
> 1. $y_c(x)$ (**Complementary Function**): The system's natural free response to initial disturbances, governed by $am^2 + bm + c = 0$.
> 2. $y_p(x)$ (**Particular Solution**): The steady-state forced response dictated by the driving force $g(x)$.

The Intuitive Physical Analogy: Pushing a Child on a Swing

Imagine pushing a child on a playground swing:

- If you let the swing go from an initial angle without pushing, it swings and naturally slows down: that is the **natural response** $y_c(x)$.

- If you continuously push the swing with a rhythmic periodic force $g(x) = F_0 \sin(\omega t)$, the swing will eventually follow the rhythm of your pushes: that is the **particular response** $y_p(x)$.
- **Resonance & The Multiplication Rule**: What happens if your push frequency exactly matches the natural frequency of the swing ($\omega = \omega_0$)? The swing does not just oscillate with constant amplitude—it builds up higher and higher with each cycle! Mathematically, the driving force duplicates a term already present in $y_c(t)$, causing the amplitude to grow by a factor of t: $y_p(t) = t(A\cos\omega t + B\sin\omega t)$.

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2. In-Depth Curriculum & Slide-by-Slide Dissection

Governing General Nonhomogeneous Linear ODE

From Slide 3, an n-th order nonhomogeneous linear ODE with constant coefficients has the form:

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = g(x)$$

where a_i are real constants and $a_n \neq 0$.

The 3-Step Master Algorithm

To find the complete general solution:

1. **Step I**: Solve the associated homogeneous equation ($g(x) = 0$) to obtain the complementary function $y_c(x) = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$.
2. **Step II**: Find any one particular solution $y_p(x)$ satisfying the nonhomogeneous equation using the **Method of Undetermined Coefficients**.
3. **Step III**: Form the complete general solution:

$$\boxed{y(x) = y_c(x) + y_p(x)}$$

(Crucial Note: Initial conditions $y(x_0) = y_0, y'(x_0) = y_1$ must ALWAYS be applied to the combined solution $y = y_c + y_p$, never to y_c alone!)

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3. The Method of Undetermined Coefficients

Scope & Admissible Driving Functions

The method of undetermined coefficients makes an educated guess for y_p based on the derivatives of $g(x)$. Because linear combinations of derivatives of polynomials, exponentials, sines, and cosines generate functions of the exact same family:

- Constants: $g(x) = k \implies y_p = A$
- Polynomials: $g(x) = a_n x^n + \cdots + a_0 \implies y_p = A_n x^n + \cdots + A_1 x + A_0$
- Exponentials: $g(x) = e^{\alpha x} \implies y_p = Ae^{\alpha x}$
- Sines/Cosines: $g(x) = \sin(\beta x)$ or $\cos(\beta x) \implies y_p = A\cos(\beta x) + B\sin(\beta x)$
- Products: $g(x) = x^k e^{\alpha x}\cos(\beta x) \implies$ product of corresponding polynomial, exponential, and trig trial forms.

Case I: The Form Rule (No Duplication)

When no term in the assumed trial form of $y_p(x)$ appears in the complementary function $y_c(x)$:

$$y_p(x) = y_{p1} + y_{p2} + \cdots + y_{pm}$$

where each y_{pi} corresponds directly to the driving term in $g(x)$.

Case II: The Multiplication Rule (Resonance / Duplication)

When any term in the assumed trial form y_{pi} is identical to a term in the complementary function $y_c(x)$ (meaning it satisfies the homogeneous equation and would simply give 0 if plugged into the left side):

$$\boxed{y_{pi} \text{ must be multiplied by } x^n}$$

where n is the smallest positive integer that eliminates all duplication with terms in $y_c(x)$.

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4. Complete Step-by-Step Solutions to All Lecture 12 Slide Examples

Example 1 (Slide 6): Polynomial Forcing Term

Problem: Solve $y'' + 4y' - 2y = 2x^2 - 3x + 6$.

- **Step 1: Complementary Solution** y_c: Auxiliary equation: $m^2 + 4m - 2 = 0$. Roots by quadratic formula:

$$m = \frac{-4 \pm \sqrt{4^2 - 4(1)(-2)}}{2} = \frac{-4 \pm \sqrt{16+8}}{2} = \frac{-4 \pm \sqrt{24}}{2} = \frac{-4 \pm 2\sqrt{6}}{2} = -2 \pm$$

$$y_c(x) = c_1 e^{(-2+\sqrt{6})x} + c_2 e^{(-2-\sqrt{6})x}$$

- **Step 2: Trial Form for** y_p: Since $g(x) = 2x^2 - 3x + 6$ is a second-degree polynomial, assume:

$$y_p = Ax^2 + Bx + C$$

No terms in y_p duplicate y_c (Case I).

- **Step 3: Differentiate y_p**:

$$y_p' = 2Ax + B$$
$$y_p'' = 2A$$

- **Step 4: Substitute into the Differential Equation**:

$$(2A) + 4(2Ax + B) - 2(Ax^2 + Bx + C) = 2x^2 - 3x + 6$$

Expand and group like powers of x:

$$(-2A)x^2 + (8A - 2B)x + (2A + 4B - 2C) = 2x^2 - 3x + 6$$

- **Step 5: Equate Coefficients**:
  - For x^2:

$$-2A = 2 \implies \boxed{A = -1}$$

  - For x^1:

$$8A - 2B = -3 \implies 8(-1) - 2B = -3 \implies -8 - 2B = -3 \implies -2B = 5$$

  - For x^0 (constants):

$$2A + 4B - 2C = 6 \implies 2(-1) + 4\left(-\frac{5}{2}\right) - 2C = 6$$
$$-2 - 10 - 2C = 6 \implies -12 - 2C = 6 \implies -2C = 18 \implies \boxed{C = -9}$$

- **Step 6: Assemble Particular and General Solutions**:

$$y_p(x) = -x^2 - \frac{5}{2}x - 9$$

$$\boxed{y(x) = c_1 e^{(-2+\sqrt{6})x} + c_2 e^{(-2-\sqrt{6})x} - x^2 - \frac{5}{2}x - 9}$$

Example 2 (Slide 7): Trigonometric Forcing Term

Problem: Find a particular solution of $y'' - y' + y = 2\sin(3x)$.

- **Step 1: Check y_c Roots**: $m^2 - m + 1 = 0 \implies m = \frac{1 \pm \sqrt{-3}}{2} = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i$. Frequency is $\frac{\sqrt{3}}{2} \neq 3$, so no duplication exists.
- **Step 2: Trial Form**: Even though $g(x)$ only contains $\sin(3x)$, its first derivative produces $\cos(3x)$. Thus, the trial form MUST contain both:

$$y_p = A\cos(3x) + B\sin(3x)$$

- **Step 3: Differentiate**:

$$y_p' = -3A\sin(3x) + 3B\cos(3x)$$
$$y_p'' = -9A\cos(3x) - 9B\sin(3x)$$

- **Step 4: Substitute into the ODE**:

$$(-9A\cos 3x - 9B\sin 3x) - (-3A\sin 3x + 3B\cos 3x) + (A\cos 3x + B\sin 3x) = 2$$

Group by $\cos(3x)$ and $\sin(3x)$:

$$(-9A - 3B + A)\cos(3x) + (-9B + 3A + B)\sin(3x) = 2\sin(3x)$$
$$(-8A - 3B)\cos(3x) + (3A - 8B)\sin(3x) = 0\cos(3x) + 2\sin(3x)$$

- **Step 5: Equate Coefficients**:

$$\begin{cases} -8A - 3B = 0 \implies B = -\frac{8}{3}A \\ 3A - 8B = 2 \end{cases}$$

Substitute B into the second equation:

$$3A - 8\left(-\frac{8}{3}A\right) = 2 \implies 3A + \frac{64}{3}A = 2$$
$$\frac{9 + 64}{3}A = 2 \implies \frac{73}{3}A = 2 \implies \boxed{A = \frac{6}{73}}$$

Solve for B:

$$B = -\frac{8}{3}\left(\frac{6}{73}\right) = -\frac{16}{73} \implies \boxed{B = -\frac{16}{73}}$$

- **Step 6: State the Particular Solution**:

$$\boxed{y_p(x) = \frac{6}{73}\cos(3x) - \frac{16}{73}\sin(3x)}$$

Example 3 (Slide 8): Sum of Polynomial and Exponential-Polynomial

Problem: Solve $y'' - 2y' - 3y = 4x - 5 + 6xe^{2x}$.

- **Step 1: Complementary Solution** y_c: Auxiliary equation: $m^2 - 2m - 3 = 0 \implies (m - 3)(m + 1) = 0 \implies m_1 = 3, m_2 = -1$.

$$y_c(x) = c_1 e^{3x} + c_2 e^{-x}$$

- **Step 2: Trial Form for** y_p: The forcing term has two parts:

1. $g_1(x) = 4x - 5 \implies y_{p1} = Ax + B$
2. $g_2(x) = 6xe^{2x} \implies y_{p2} = (Cx + D)e^{2x}$ Comparing with $y_c = c_1 e^{3x} + c_2 e^{-x}$, neither $1, x$ nor e^{2x}, xe^{2x} duplicate terms in y_c.

$$y_p = Ax + B + (Cx + D)e^{2x}$$

- **Step 3: Differentiate**:

$$y_p' = A + Ce^{2x} + 2(Cx + D)e^{2x} = A + (2Cx + C + 2D)e^{2x}$$
$$y_p'' = 2Ce^{2x} + 2(2Cx + C + 2D)e^{2x} = (4Cx + 4C + 4D)e^{2x}$$

- **Step 4: Substitute and Group**:
  - Polynomial part:

$$-2(A) - 3(Ax + B) = 4x - 5 \implies -3Ax - (2A + 3B) = 4x - 5$$

$$-3A = 4 \implies \boxed{A = -\frac{4}{3}}$$

$$-(2A + 3B) = -5 \implies 2\left(-\frac{4}{3}\right) + 3B = 5 \implies -\frac{8}{3} + 3B = 5 \implies 3B =$$

  - Exponential part:

$$(4Cx + 4C + 4D)e^{2x} - 2(2Cx + C + 2D)e^{2x} - 3(Cx + D)e^{2x} = 6xe^{2x}$$

Divide by e^{2x}:

$$(4C - 4C - 3C)x + (4C + 4D - 2C - 4D - 3D) = 6x$$
$$-3Cx + (2C - 3D) = 6x + 0$$
$$-3C = 6 \implies \boxed{C = -2}$$

$$2C - 3D = 0 \implies 2(-2) - 3D = 0 \implies 3D = -4 \implies \boxed{D = -\frac{4}{3}}$$

- **Step 5: Assemble General Solution**:

$$\boxed{y(x) = c_1 e^{3x} + c_2 e^{-x} - \frac{4}{3}x + \frac{23}{9} - \left(2x + \frac{4}{3}\right) e^{2x}}$$

Example 4 (Slide 9): Case II Duplication with Single Exponential

Problem: Find a particular solution of $y'' - 5y' + 4y = 8e^x$.

- **Step 1: Complementary Solution** y_c:

$$m^2 - 5m + 4 = 0 \implies (m-1)(m-4) = 0 \implies m_1 = 1, m_2 = 4$$
$$y_c(x) = c_1 e^x + c_2 e^{4x}$$

- **Step 2: Detect Duplication and Apply Multiplication Rule**: Naive guess: $y_p = Ae^x$. **CRITICAL TRAP**: e^x is already in $y_c(x)$! If we plug Ae^x into the left-hand side, it will equal 0, not $8e^x$! Multiply by x^1:

$$y_p = Axe^x$$

  Check: xe^x is not in y_c. Duplication is eliminated!

- **Step 3: Differentiate** y_p:

$$y_p' = Ae^x + Axe^x = A(x+1)e^x$$
$$y_p'' = Ae^x + A(x+1)e^x = A(x+2)e^x$$

- **Step 4: Substitute into the ODE**:

$$A(x+2)e^x - 5A(x+1)e^x + 4Axe^x = 8e^x$$

  Divide by e^x:

$$A(x+2) - 5A(x+1) + 4Ax = 8$$
$$(A - 5A + 4A)x + (2A - 5A) = 8$$
$$0x - 3A = 8 \implies -3A = 8 \implies \boxed{A = -\frac{8}{3}}$$

- **Step 5: State the Verified Particular Solution**:

$$\boxed{y_p(x) = -\frac{8}{3}xe^x}$$

Example 8 (Slide 14): Case II with Repeated Root in y_c

Problem: Find a particular solution of $y'' - 2y' + y = e^x$.

- **Step 1: Complementary Solution** y_c:

$$m^2 - 2m + 1 = 0 \implies (m-1)^2 = 0 \implies m_1 = m_2 = 1$$
$$y_c(x) = c_1 e^x + c_2 x e^x$$

- **Step 2: Detect Duplication**:
  - Guess 1: $y_p = Ae^x \implies$ duplicates $c_1 e^x$.
  - Guess 2: Multiply by $x \implies y_p = Axe^x \implies$ duplicates $c_2 x e^x$!
  - Guess 3: Multiply by $x^2 \implies y_p = Ax^2 e^x$. $x^2 e^x$ is NOT in y_c, so multiplication rule index is $n = 2$!

- **Step 3: Differentiate** $y_p = Ax^2 e^x$:

$$y_p' = 2Axe^x + Ax^2 e^x = A(x^2 + 2x)e^x$$
$$y_p'' = A(2x+2)e^x + A(x^2+2x)e^x = A(x^2 + 4x + 2)e^x$$

- **Step 4: Substitute into the ODE**:

$$A(x^2+4x+2)e^x - 2A(x^2+2x)e^x + Ax^2 e^x = e^x$$

Divide by e^x:

$$A(x^2+4x+2) - 2A(x^2+2x) + Ax^2 = 1$$
$$(A - 2A + A)x^2 + (4A - 4A)x + 2A = 1$$
$$0x^2 + 0x + 2A = 1 \implies 2A = 1 \implies \boxed{A = \frac{1}{2}}$$

- **Step 5: Particular Solution**:

$$\boxed{y_p(x) = \frac{1}{2}x^2 e^x}$$

Example 11 (Slide 18): Resonance Initial-Value Problem

Problem: Solve the initial-value problem:

$$y'' + y = 4x + 10\sin x, \qquad y(\pi) = 0, \quad y'(\pi) = 2$$

- **Step 1: Complementary Solution y_c**:

$$m^2 + 1 = 0 \implies m = \pm i \implies y_c(x) = c_1 \cos x + c_2 \sin x$$

- **Step 2: Trial Form for y_p**:
  - For $4x$: $y_{p1} = Ax + B$ (no duplication).
  - For $10 \sin x$: standard guess is $C \cos x + D \sin x$, but both $\cos x$ and $\sin x$ are in y_c! Multiply by x: $y_{p2} = x(C \cos x + D \sin x) = Cx \cos x + Dx \sin x$. Combined trial form:

$$y_p = Ax + B + Cx \cos x + Dx \sin x$$

- **Step 3: Solve for Coefficients**:
  - For $y_{p1} = Ax + B$:

$$y_{p1}'' + y_{p1} = 0 + (Ax + B) = 4x \implies \boxed{A = 4}, \quad \boxed{B = 0}$$

  - For $y_{p2} = Cx \cos x + Dx \sin x$:

$$y_{p2}' = C \cos x - Cx \sin x + D \sin x + Dx \cos x = (C + Dx) \cos x + (D - Cx)$$

$$y_{p2}'' = -C \sin x + D \cos x - Dx \sin x + D \cos x - C \sin x - Cx \cos x$$

$$y_{p2}'' = (2D - Cx) \cos x + (-2C - Dx) \sin x$$

Substitute into $y'' + y$:

$$(2D - Cx) \cos x + (-2C - Dx) \sin x + (Cx \cos x + Dx \sin x) = 10 \sin x$$

$$2D \cos x - 2C \sin x = 10 \sin x$$

Equate coefficients:

$$2D = 0 \implies \boxed{D = 0}$$

$$-2C = 10 \implies \boxed{C = -5}$$

Thus:

$$y_p(x) = 4x - 5x \cos x$$

- **Step 4: Form General Solution**:

$$y(x) = c_1 \cos x + c_2 \sin x + 4x - 5x \cos x$$

- **Step 5: Apply Initial Conditions at $x = \pi$**:

- $y(\pi) = 0$:

$$c_1 \cos(\pi) + c_2 \sin(\pi) + 4\pi - 5\pi \cos(\pi) = 0$$

$$c_1(-1) + c_2(0) + 4\pi - 5\pi(-1) = 0$$

$$-c_1 + 4\pi + 5\pi = 0 \implies -c_1 + 9\pi = 0 \implies \boxed{c_1 = 9\pi}$$

- Differentiate $y(x)$:

$$y'(x) = -c_1 \sin x + c_2 \cos x + 4 - 5(\cos x - x \sin x) = -c_1 \sin x + (c_2 - 5) \cos$$

- $y'(\pi) = 2$:

$$-c_1 \sin(\pi) + (c_2 - 5) \cos(\pi) + 4 + 5\pi \sin(\pi) = 2$$

$$0 + (c_2 - 5)(-1) + 4 + 0 = 2$$

$$-(c_2 - 5) + 4 = 2 \implies -c_2 + 5 + 4 = 2 \implies -c_2 + 9 = 2 \implies \boxed{c_2 = 7}$$

- **Step 6: State the Unique Solution**:

$$\boxed{y(x) = 9\pi \cos x + 7 \sin x + 4x - 5x \cos x}$$

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5. Common Exam Pitfalls & Trap Analysis

Exam Pitfall / Trap	Why Students Make This Mistake	How the Exam Sets the Trap	The Bulletproof Defense
Solving for c_1, c_2 Before Finding y_p	Instinctively applying initial conditions to y_c as soon as the auxiliary roots are found	The exam gives an IVP ($y(0) = y_0, y'(0) = y_1$); student applies it to y_c, then adds y_p	**GOLDEN RULE**: Constants c_1, c_2 can ONLY be determined from the TOTAL solution $y = y_c + y_p$.
Omitting Cosine When Forcing Term Is Sine	Assuming $g(x) = \sin(\omega x)$ only needs $y_p = A \sin(\omega x)$	Derivative of sin produces cos, leaving unpaired terms on the left side	Always include both: $A \cos(\beta x) + B \sin(\beta x)$.
Missing Duplication with y_c	Forgetting to find y_c before guessing y_p	When $g(x) = e^x$ and $m = 1$ is a root of the auxiliary equation	Always find y_c FIRST! Compare every term of your trial form against y_c. If identical, multiply by x.
Failing to Multiply ALL	When multiplying by x for duplication, only	If $g(x) = xe^x$ and $e^x \in y_c$, guessing $(Ax + B)e^x$	Multiply the ENTIRE trial block: $x(Ax + B)e^x =$

Exam Pitfall / Trap	Why Students Make This Mistake	How the Exam Sets the Trap	The Bulletproof Defense
Parts of a Product	multiplying one term	instead of $x(Ax+B)e^x$	$(Ax^2+Bx)e^x$.
Resonance Amplitude Calculation Slip	In IVPs with $x\cos x$, forgetting product rule when evaluating $y'(x)$	Forgets the derivative of x in $\frac{d}{dx}[x\cos x] = \cos x - x\sin x$	Write each derivative line-by-line using $(uv)' = u'v + uv'$ before setting $x = x_0$.

6. Self-Check Mastery Practice Problems

Problem 1 (Polynomial & Resonance)

Find the general solution of $y'' - 4y = 8e^{2x} + 4x$.

- **Complementary**: $m^2 - 4 = 0 \implies m = \pm 2 \implies y_c = c_1e^{2x} + c_2e^{-2x}$.
- **Forcing 1 ($4x$)**: $y_{p1} = Ax + B \implies -4(Ax+B) = 4x \implies A = -1, B = 0 \implies y_{p1} = -x$.
- **Forcing 2 ($8e^{2x}$)**: e^{2x} duplicates $c_1e^{2x} \implies y_{p2} = Cxe^{2x}$. $y'_{p2} = C(1+2x)e^{2x}$, $\quad y''_{p2} = C(4+4x)e^{2x}$. $y'' - 4y = C(4+4x)e^{2x} - 4Cxe^{2x} = 4Ce^{2x} = 8e^{2x} \implies C = 2$.
- **General Solution**:

$$\boxed{y(x) = c_1e^{2x} + c_2e^{-2x} + 2xe^{2x} - x}$$